

**Q1.**

Find the general solution of the differential equation

$$\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 10y = e^{2t}$$

giving your answer in the form  $y = f(t)$ .

(Total 6 marks)

**Q2.**

- (a) Find the values of the constants  $p$  and  $q$  for which  $p + qxe^{-2x}$  is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 4 - 9e^{-2x}$$

(5)

- (b) Hence find the general solution of this differential equation.

(3)

- (c) Hence express  $y$  in terms of  $x$ , given that  $y = 4$  when  $x = 0$  and that  $\frac{dy}{dx} \rightarrow 0$  as  $x \rightarrow \infty$ .

(4)

(Total 12 marks)

**Q3.**

A particle  $P$ , of mass  $m$  kg, moves in a straight horizontal line. At time  $t$  seconds, the displacement of  $P$  from a fixed point  $O$  on the line is  $x$  metres, and  $P$  is moving with velocity  $\dot{x}$  m s<sup>-1</sup>. Throughout the motion, two horizontal forces act on  $P$ : a force of magnitude  $4mn^2|x|$  newtons directed towards  $O$ , and a resistance force of magnitude  $2mk|\dot{x}|$  newtons, where  $n$  and  $k$  are positive constants.

- (a) Show that  $\ddot{x} + 2k\dot{x} + 4n^2x = 0$ .

(3)

- (b) In one case,  $k = n$ . When  $t = 0$ ,  $x = a$  and  $\dot{x} = 0$ .

(i) Show that  $x = e^{-nt} \left( a \cos \sqrt{3}nt + \left( \frac{\sqrt{3}a}{3} \right) \sin \sqrt{3}nt \right)$ .

(7)

- (ii) Show that  $P$  passes through  $O$  when  $\tan \sqrt{3}nt = -\sqrt{3}$ .

(3)

- (c) In a different case,  $k = 2n$ .

- (i) Find a general solution for  $x$  at time  $t$  seconds.

(3)

- (ii) Hence state the type of damping which occurs.

(1)

(Total 17 marks)

#### Q4.

A particle moves in a straight line with simple harmonic motion such that its displacement at time  $t$  seconds relative to a fixed origin on this line is  $x$  metres. The motion of the particle satisfies the differential equation

$$\frac{d^2 x}{dt^2} + 16x = 0$$

- (a) Verify that

$$x = A \cos 4t + B \sin 4t$$

where  $A$  and  $B$  are constants, is a solution to this differential equation.

(4)

- (b) When  $t = 0$ , the particle is momentarily at rest. Show that  $B = 0$ .

(1)

- (c) Given that  $x = h$  ( $h > 0$ ) when  $t = \frac{\pi}{12}$ , find  $A$  in terms of  $h$ .

(2)

- (d) Find the maximum speed of the particle in terms of  $h$ .

(1)

- (e) The mass of the particle is  $m$  kg. Find the magnitude of the maximum force acting on the particle during the motion. Give your answer in terms of  $h$  and  $m$ .

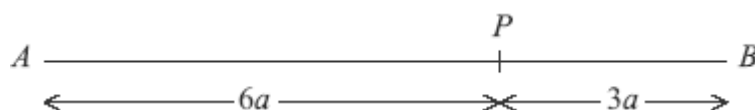
(2)

(Total 10 marks)

#### Q5.

The ends of a light, uniform elastic string are fixed to two points,  $A$  and  $B$ , a distance  $9a$  apart on a smooth, horizontal plane. The string is of natural length  $6a$  and modulus of elasticity  $4mn^2a$ , where  $n$  is a constant.

A particle of mass  $m$  is attached to the string at  $P$ , where  $AP = 6a$ . The natural length of  $AP$  is  $4a$  and the natural length of  $BP$  is  $2a$ . In this position, the particle is in equilibrium.



The particle is moved a distance  $\frac{1}{2}a$  towards  $B$  and then released from rest at time  $t = 0$ . The displacement of the particle from its equilibrium position at time  $t$  is  $x$ .

Hence initially  $x = +\frac{1}{2}a$ .

The motion of the particle is resisted by a force of magnitude  $2mnv$ , where  $v$  is the speed of the particle at time  $t$ .

- (a) Show that  $x$  satisfies

$$\frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 3n^2x = 0$$

(7)

- (b) Given that  $n = 1$ , find  $x$  in terms of  $a$  and  $t$ .

(8)

(Total 15 marks)

**Q6.**

A sphere of mass 2 kg is fixed to one end of a spring. The other end of the spring is attached to the fixed point  $P$ , as shown in the diagram. The natural length of the spring is 0.5 metres.



- (a) When the system is in equilibrium, the length of the spring is 0.7 metres.

Find the modulus of elasticity of the spring.

(3)

- (b) The sphere is pulled down below its equilibrium position and released from rest. At time  $t$  seconds, the displacement of the sphere from its equilibrium position is  $x$  metres.

- (i) Show that

$$\frac{d^2x}{dt^2} = -49x$$

(5)

- (ii) Find the value of  $t$  when the sphere returns to its equilibrium position for the first time.

(4)  
(Total 12 marks)

**Q7.**

An elastic string has natural length  $l$  and modulus of elasticity  $\lambda$ . One end of the string is fixed and a particle of mass  $m$  is attached to the other end of the string.

- (a) When the particle hangs vertically in equilibrium, the extension of the string is  $\frac{l}{4}$ . Show that  $\lambda = 4mg$ .

(2)

- (b) The particle is then set into vertical motion and  $x$  is the displacement of the particle from the equilibrium position at time  $t$ . Assume that the string remains taut during the vertical motion.

(i) Show that  $\frac{d^2 x}{dt^2} = \frac{4gx}{l}$ .

(5)

- (ii) Hence find the period of the motion.

(2)

(Total 9 marks)

**Q8.**

A spring, of natural length  $4a$  and modulus  $\lambda$ , has one end attached to a fixed support  $A$ , and a particle  $P$  of mass  $m$  is attached to its other end. Another spring, of natural length  $2a$  and modulus  $4mg$ , has one end attached to  $P$  and the other end attached to a fixed support  $B$ , which is situated at a distance of  $10a$  vertically below  $A$ . The system is in equilibrium in a vertical line with the upper spring stretched to a length of  $7a$  and the lower spring stretched to a length of  $3a$  as shown in the diagram.



- (a) Show that  $\lambda = 4mg$ .

(4)

- (b) At time  $t = 0$ , the particle is lowered to a distance  $\frac{a}{2}$  below its equilibrium position and released from rest. The subsequent motion of  $P$  is subject to a resistance of

magnitude  $\frac{1}{5}mkv$ , where  $k^2 = \frac{6g}{a}$  and  $v$  is the speed of the particle at time  $t$ .

- (i) Given that  $x$  is the downward displacement of  $P$  from its equilibrium position at time  $t$ , show that

$$10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5k^2x = 0 \quad (6)$$

- (ii) Hence find  $x$  in terms of  $a$ ,  $k$  and  $t$ .

(9)

(Total 19 marks)

### Q9.

A particle, of mass  $m$ , is attached to one end of a spring. The spring has natural length  $l$  and modulus of elasticity  $\lambda$ . The other end of the spring is fixed to the point  $Q$ . Initially the particle is in equilibrium, vertically below  $Q$ .

- (a) Find, in terms of  $m$ ,  $g$ ,  $\lambda$  and  $l$ , the extension of the spring when the particle is in equilibrium. (2)
- (b) The particle is then pulled down, so that it is vertically below its equilibrium position, and released. The displacement of the particle from its equilibrium position is  $x$  at time  $t$ . Show that

$$\frac{d^2x}{dt^2} = -\frac{\lambda x}{ml} \quad (5)$$

- (c) Find, in terms of  $\lambda$ ,  $l$  and  $m$ , the period of the subsequent motion. (2)
- (d) The spring is now replaced with another spring of natural length  $l$ , but with modulus of elasticity  $\frac{\lambda}{4}$ . What happens to the period of the motion? (2)

(Total 11 marks)

### Q10.

A particle moves on a straight line. At time  $t$  the displacement of the particle from a point  $O$  is  $x$ . The particle moves with simple harmonic motion, so that

$$\frac{d^2x}{dt^2} = -25x$$

The particle is initially at rest at the point  $P$ , which is a distance  $a$  from  $O$ .

- (a) Find the time that it takes the particle to move directly from  $P$  to  $O$ .

(3)

- (b) Find, in terms of  $a$ , the distance of the particle from  $O$  when its speed is half its maximum speed.

(5)

(Total 8 marks)

**Q11.**

The motion of a particle, which moves along a straight line, is governed by the differential equation

$$2 \frac{d^2 x}{dt^2} + 6 \frac{dx}{dt} + 9x = 0$$

where  $x$  is the displacement of the particle from the origin at time  $t$ .

Initially the particle is at rest at the point where  $x = a$ .

- (a) Find  $x$  in terms of  $a$  and  $t$ .

(9)

- (b) Is the damping of the motion of the particle light, critical or heavy? Give a reason for your answer.

(2)

(Total 11 marks)